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Sustainable Development Goals enabled by Additive Manufacturing: a design perspective

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Abstract. The human exploitation of resources is currently exceeding the real possibilities of the planet. New sustainable models of consumption and production have been implemented in the last years to limit the impact of human activities, i.e., the definition of the 17 Sustainable Development Goals (SDGs) in 2015. Design and digital technology can potentially foster the achievement of these goals, although their real contribution to this transition has not been defined yet. This paper aims to investigate the link between additive manufacturing technologies and design for sustainability for the achievement of the 17 United Nations SDGs. After a preliminary review of the goals and the theoretical framework, the most relevant case studies from the practical context were selected to deepen the investigation. Three matrixes were then created to detect the possible interconnections with the SDGs and the key aspects of the theoretical framework. Potential links were found with 15 of the 17 SDGs. Accessible additive manufacturing technologies can foster sustainable models, especially within local scales and communities. Designers can be seen as facilitators or creators of this transition from a practical point of view, fostering behavioral change towards more sustainable models.

Keywords: Additive Manufacturing, 3D Printing, Design for Sustainability, Circular Economy, Sustainable Development Goals (SDGs).

1 Introduction

Nowadays, human activities strongly affect the global environment at different levels and, consequently, the future of the planet. Due to the noticeable impact of the human-induced changes, the current geological era has been renamed the “Anthropocene” [1]. One warning alarm is represented by the increase of the material output from human activities, namely the “anthropogenic mass”, which will exceed the living mass present on Earth in the next few years [2]. It means that new models

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of sustainable production and consumption have to be investigated in the short and mid-term since the current exploitation of natural resources is overcoming the real possibilities of the planet. For these reasons, the concepts of sustainability, sustainable development, and circular economy have been assuming a crucial role for a broader audience of stakeholders that aim to reach the transition toward more sustainable models such as industries, researchers, and policy-makers [3–5]. New sustainable practices and initiatives have been encouraged and implemented in the last years. In 2015, the United Nations (UN) defined 17 Sustainable Development Goals (SDGs) to be achieved by 2030. These goals were meant to facilitate the transition toward more sustainable paradigms from an integrated perspective involving the environmental, social, and economic dimensions. In other words, UN SDGs defined some practical targets to be globally reached by the next decade [6, 7].

Within this context, the design discipline represents a key aspect of facilitating the sustainable transition. In particular, the research area of Design for Sustainability aims to increase the awareness of the new generation of designers on these topics, developing new connections and transdisciplinary tools for the practitioners [8, 9].

At the same time, digital technologies have shown the potential to significantly contribute to this transition and the achievement of the SDGs [10–12]. In particular, Additive Manufacturing (AM), also known as 3D printing, seems to foster the development of new strategies that involve sustainability and circular economy since its use has been gradually changing through time. As a matter of fact, 3D printing is increasingly considered a proper process available for the users rather than a prototyping tool for generating new products [13–17]. Despite this current situation, the role of AM in the achievement of the SDGs has not been clearly defined yet, especially considering its potential links with the field of Design for Sustainability. Only a few works focused on the potentials represented by AM at the intersection with design and circular economy [18, 19].

From literature, the development and the use of new materials also influence the transition toward more sustainable models of production and consumption. New hybrid practices and professional figures have emerged to implement sustainable models, namely the Do-It-Yourself (DIY) materials approach and the material designers, respectively [20, 21]. In the light of the above, new synergies can be detected at the intersection of design, materials, and AM for new circular economy models [22], but there is no clear evidence of their contribution to the achievement of the SDGs.

This paper focuses on the interconnection between AM and design for sustainability for the transition toward more sustainable ways of living from an environmental, social, and economic point of view. In particular, the goal is to review and examine the role of accessible AM technologies in achieving the 17 United Nations (UN) Sustainable Development Goals (SDGs). Afterward, it also detects the main aspects of AM technologies related to the SDGs by analyzing the most relevant case studies from the practical context.

This paper mainly addresses the following three Research Questions (RQs):

1. How can AM technologies, especially accessible and low-cost 3D printing, foster the SDGs?
2. Which are the main aspects related to AM that contribute to achieving these SDGs?
3. What is design practice's role in this sustainable transition enabled by accessible and low-cost AM?

After preliminary literature on the theoretical framework of this topic, the UN SDGs were analyzed considering the potential links with AM. A total of 8 key aspects were defined at the intersection of AM and the SDGs, and they helped deepen the analysis through selected practice-based case studies from 2015 to 2021. Three different matrixes were then created to highlight the possible interconnections between the selected case studies, the SDGs, and the 8 key aspects to address the RQs. Possible interconnections were found with 15 of the 17 UN SDGs, and extrusion-based AM processes appear to be the most appropriate 3D printing technologies that foster sustainable transitions, especially low-cost 3D printers. As a matter of fact, a larger number of bottom-up initiatives and local scale closed loops have been implementing sustainable and circular practices. In the next few years, new strategies can be implemented to promote the achievement of the SDGs, and design practice can play an active role in fostering behavioral change towards more sustainable ways of living.

2 Research methods

This work aims to better understand the role of design practice and accessible AM technologies to achieve the SDGs by reviewing the main practice-based case studies. A preliminary literature review was conducted on Scopus to assume the theoretical framework related to the interconnection between AM and design for sustainability. The 17 UN SDGs were then analyzed considering the potential contribution of AM.

In detail, the analysis was focused on the specific targets and indicators that AM can influence, and the previous literature review helped in detecting the main concepts related to the SDGs. The SDGs were also divided into three main clusters according to the contribution of the Stockholm Resilience Centre (SRC). From this analysis, the goals can be related to the Biosphere, the Society, and the Economy, and these three aspects should be seen as embedded parts where the Biosphere includes the Society, which in turn includes the Economy [6, 7]. These aspects were then linked with the SDGs and the analysis of this paper. These two steps were also helpful in defining 8 key aspects related to AM that mainly contribute to SDGs achievement. More in detail, these key concepts can be defined as transdisciplinary aspects where AM plays a crucial role in the practical transition toward more sustainable models. Further details are described in the next section.

Afterward, a deeper analysis was done by selecting the most relevant case studies within the practical context from 2015 to 2021, which means from the designers' and practitioners' ongoing projects and activities. Only works published from 2015 were considered because the UN SDGs were in 2015 for the first time in 2015 [6, 7]. To

this end, the main repositories and blogs related to design practice and 3D printing technologies were searched instead of scholarly databases, i.e., 3dprint.com; 3dprintingindustry.com; dezeen.com; designboom.com. As a matter of fact, practice-based projects can be easily explored within those kinds of repositories. After a first screening to remove the duplicates, the link with at least one of the 8 key aspects and the use of extrusion-based AM technologies were considered the main eligibility criteria. The latter aspect derives from the preliminary literature review of the theoretical framework that showed this AM category as the most common to foster the transition towards sustainable ways of production and consumption. After detecting 31 case studies, a deeper analysis was then performed by checking the completeness of the information on the websites of the specific case study, i.e., the project or the partner websites. Works with a clear link with at least four of the 8 key aspects were considered for the review, while projects with weak or unclear connections were discarded, such as not enough information or theoretical concepts and/or proof-of-concept.

In the end, a total of 10 case studies were collected for the analysis. The 17 UN SDGs, the 8 key aspects, and the 10 selected case studies were then described and combined to create three different matrixes for the data analysis. In particular, these matrixes aim to underline the interconnections between:

1. The 10 selected case studies and the 8 key aspects from the preliminary review (matrix 1).
2. The 10 selected case studies and the 17 UN SDGs (matrix 2).
3. The 8 key aspects (preliminary literature review) and the 17 UN SDGs (matrix 3).

The possible interconnections between the rows and the columns were marked by X. The three columns with the highest number of X for each matrix were highlighted to indicate the most frequent interconnections derived from the analysis, i.e., the 3 key aspects or 3 UN SDGs according to the specific case. In the end, these results were discussed to address the RQs considering the three dimensions of sustainable development, and some future perspectives were briefly depicted.

3 Results and discussion

3.1 SDGs analysis

The general analysis of the SDGs aimed to highlight the potential contribution of design practice and AM technologies in fostering their achievement. In detail, possible interconnections were detected by studying the specific targets of the 17 goals. In the beginning, these potential links were expressed through some keywords. Afterward, a total of 8 key aspects were defined thanks to the preliminary literature review. In general, accessible AM technologies can be linked with the UN SDGs, especially considering the activities related to the design practice [6, 7]. Possible interconnections were therefore found for 15 of the 17 SDGs:

- **SDG 1** (No Poverty) aims to eradicate poverty situations, ensure social protection measures, and equal rights, reduce exposure to extreme events, and support policy frameworks for new development strategies. Within its 7 targets, AM and design practice can mainly contribute to targets 1.4 and 1.5 through concrete local actions for the democratization of technology and open and frugal innovation.
- **SDG 2** (Zero Hunger) wants to end hunger and malnutrition, promote sustainable and equitable agriculture, and increase investments and food security. Targets 2.4 and 2.5 can be fostered thanks to AM and design practice by promoting the use of technology in rural and local communities.
- **SDG 3** (Good Health and Well-being) focuses on ensuring healthy lives amongst the individuals by reducing mortality ratios and epidemics, promoting mental health and well-being, and supporting the research of new treatments. 3 of the 13 indicators (3.3, 3.4, and 3.8) could be supported by AM and design through customized and distributed assistive or medical solutions and bioprinting technologies.
- **SDG 4** (Quality Education) is related to inclusive and equitable quality lifelong education avoiding discrimination. AM and design practices could enhance targets from 4.1 to 4.5 and 4.a to open innovation principles and accessible facilities, i.e., makerspaces.
- **SDG 5** (Gender Equality) aims to empower vulnerable subjects through participation and equal opportunities. Target 5.b seems to be particularly connected with AM and design practice since accessible and open technology could be a powerful way for empowerment.
- **SDG 6** (Clean Water and Sanitation) ensures sustainable accessibility of water and sanitation considering vulnerable situations, water-use efficiency, and technologies for waste treatment and ecosystem restoration. To this end, targets from 6.6 to 6.a focus on the use of strategies and technologies, including the contribution of AM and design practice.
- **SDG 7** (Affordable and Clean Energy) wants to give access to affordable energy, especially from renewable sources and new technological progress. The contribution from AM and design practice seems to be limited considering these topics.
- **SDG 8** (Decent Work and Economic Growth) promotes sustainable economic growth and employment through innovation, creativity, technological upgrading, resource efficiency, equity, and safety. Within its 12 targets, AM and design can contribute to targets from 8.2 to 8.6 and 8.9 through local actions for accessible technology, distributed manufacturing, and open innovation.
- **SDG 9** (Industry, Innovation, and Infrastructure) wants to create resilient, inclusive, and sustainable infrastructures for innovation by supporting technological progress, accessible communication, and small-scale enterprises. AM and design can strongly contribute to targets from 9.2 to 9.5, 9.b, and 9.c through distributed manufacturing, open innovation, and circular economy principles.
- **SDG 10** (Reduced Inequalities) aims to reduce inequality at different levels. Within its 10 targets, design practice and AM may foster the inclusiveness of

individuals of target 10.2 through accessible technologies and customizable solutions.

- **SDG 11** (Sustainable Cities and Communities) wants to create inclusive, resilient, and sustainable human settlements by ensuring access to essential services and enhancing participation, integration, and resilience in local communities. Targets 11.1, 11.3, 11.4, 11.6, 11.7, 11.b and 11.c can be influenced by design practice and AM technologies through local-scale actions, accessible technology, and open innovation.
- **SDG 12** (Responsible Consumption and Production) fosters sustainable consumption and production models through efficient management of resources and wastes, awareness of sustainable lifestyles, and technological progress. Targets 12.2, 12.4, 12.5, 12.8, 12.a, and 12.b can be supported by circular design practices and AM technologies in distributed manufacturing contexts.
- **SDG 13** (Climate Action) wants to contrast the impacts of climate change through new strategic policies and awareness-raising actions. Target 13.1 can be supported by resilient and circular local activities.
- **SDG 14** (Life below Water) aims for the sustainable use of marine resources by reducing pollution and protecting marine ecosystems. Design practices and AM technology can contribute to Targets 14.1 and 14.2 through circular economy practices and ecosystem restoration strategies.
- **SDG 15** (Life on Land) is meant to protect and restore terrestrial ecosystems and biodiversity through concrete actions and prevention. Targets 15.1, 15.2, and 15.5. can be linked to AM and design practices through circular economy principles and open innovation.
- **SDG 16** (Peace, Justice, and Strong Institutions) promotes inclusive societies and accountable institutions. Due to the extension of this topic, the contribution of low-cost AM combined with design practices seems limited at the moment.
- **SDG 17** (Partnerships for the Goals) aims to implement the global partnership for sustainable development. Targets from 17.6 to 17.8 can be linked to global and local actions for accessible technology and open solutions.

In addition, the UN SDGs can be divided according to the environmental aspect (Biosphere, SDGs 6, 13, 14, and 15), the social aspect (Society, SDGs 1, 2, 3, 4, 5, 7, 11, and 16) and the economic aspect (Economy, SDGs 8, 9, 10, and 12) as embedded clusters of sustainable development [6, 7]. SDG 17 could be set apart from these three clusters since it is not focused on a specific aspect and mainly encourages global actions towards more sustainable models.

3.2 Key interconnecting aspects between AM, design practice, and SDGs

After the previous analysis, practical interconnections toward more sustainable models were translated into 8 key aspects. The keywords from the possible connections were further defined through the preliminary literature review, resulting in the following key aspects:

1. **Democratization of technology, open and frugal innovation** as vehicles of inclusive and participative knowledge sharing processes [23–25].
2. **Personal fabrication and Do-It-Yourself (DIY) practices** for “on-demand” products and materials [17, 21].
3. Distributed Manufacturing, local scales, and **communities** for the creation of global sustainable networks of local production communities [26–28].
4. **Assistive technology, bioprinting, medicine** to fulfill user-specific needs [29].
5. **Mass customization** for personalized products [30, 31].
6. **The spread of fablabs/makerspaces** for global open networks of exchange [27].
7. **Circular economy enabled by AM** according to the implementation strategies based on the R-imperatives, i.e. Recycling, Reuse, Repair, and Repurpose [22].
8. **Ecosystem restoration and preservation through AM** as an open and accessible tool [32, 33].

From the literature review, extrusion-based AM seems to be more used than the other 3d printing processes to create new design projects and strategies for achieving the SDGs, i.e. some FFF (Fused Filament Fabrication, also known as Fused Deposition Modelling) printers [34]. This category (“Material Extrusion” AM) is based on the extrusion of a specific material through a nozzle that selectively reproduces the desired shape layer-by-layer [35]. Its spread in the sustainable transition can be related to several reasons such as: (i) its capability to foster new DIY and bottom-up initiatives from local scales and communities, (ii) the reduced costs compared to other AM technologies while allowing to achieve complex customizable shapes, (iii) the possibility to use local resources and places for new closed loops, and (iv) the implementation of new circular economy strategies. Despite other projects involving different AM technologies linked with the SDGs, the analysis focused on the accessible and low-cost extrusion-based AM since its potential can be related to a more significant number of SDGs, i.e., democratizing this technology for a broader audience. Designers may assume a crucial role in facilitating the sustainable transition by fostering the use of this AM technology within their professional activity, starting its exploitation.

3.3 Practice-based case studies

A total of 10 practice-based case studies were detected considering the SDGs and the 8 key aspects explained in the previous paragraph. Table 1 collects the selected practice-based case studies from 2015 to 2021 following the chronological order and resumes their main details such as year, name, partners, and brief description. Despite the limited number of case studies, an increasing interest in sustainability-related topics can be noticed in the last two years since half of the case studies started in 2020 and 2021. Some case studies have been currently going on with their primary activities, i.e., n. 2, 6, 9, 10, whereas other ended projects were part of wider strategies or entrepreneurship visions, i.e., 1, 3, 4, 7, 8. Hence, the different stakeholders seem to be aware of the necessity to plan mid-and long-term actions to foster the transition towards sustainable models of production and consumption. Also,

Table 1 shows that several actors are involved in encouraging this behavioral change, and it can be done at different levels. Some projects involved partners from the academic and industrial contexts (n. 3, 4, 5, 6, 10) or even volunteers and end-users (2, 9).

On the one hand, industries and design practitioners appear as the main actors that can spread concrete practices to achieve SDGs, especially at local levels. On the other hand, global actions can be encouraged by creating virtual networks of interested stakeholders, including non-design professionals and volunteer figures (n. 2, 3, 9). For this reason, matching global intangible networks and local practical actions could be an interesting way to spread new ways of living towards the perspective of the SDGs. Finally, the heterogeneity of the case studies demonstrates that different approaches can be followed to achieve these goals thanks to the intrinsic flexibility of both design practice and AM technologies.

Table 1. Year, project name, partners, and brief descriptions of the practice-based case studies described in this work.

N.	Year	Project name	Partners	Project description
1	2016	“Maker economy starter kit”	WASP S.r.l.	Container kit for local and open source self-production of 3D printed products [36];
2	2016	“E-Nable”	Volunteers network (makers, designers, engineers)	Global network for the local production of accessible 3D printed assistive technology devices [37];
3	2018	“One Hand for Syria”	WASP Hub Mantova, AMAR, Creative Lab, Damascus University, Fab Lab Mantova, Helpy Technology, NUR, Mondoinsieme Foundation	Local manufacturing of 3D printed free prosthetic limbs for Damascus war mutilates [38];
4	2018	“Kijenzi”	Michigan Tech’s Open Sustainability Technology Lab, Pennsylvania State University	Development of an open-source 3D printer for humanitarian crisis response [39];
5	2019	“H.O.R.T.U.S . XL”	ecoLogicStudio, CREATE Group, WASP Hub Denmark, University of Southern Denmark, YIP Structural Engineering, Ecoduna AG, Extrudr, Synthetic Landscape Lab	Development of 3D printed bio-digital architectures for oxygen and biomass conversion [40];
6	2020	“Hiveopolis”	University of Graz, EPFL, Université libre de Bruxelles, Frei Universität Berlin, Pollenity, Latvia University of Life Sciences & Technologies, Humboldt-Universität zu Berlin	Development of 3D printed beehives for cooperation between humans and bee colonies [41];

7	2020	“Community First! Village”	Mobile Loaves & Fishes nonprofit foundation, ICON Technology Inc.	3D printed welcome home structures for homeless people [42];
8	2021	“The Elements”	The New Raw, Coca-Cola (Greece), Ogilvy Greece, EcoRec	New 3D printed furniture from upcycled marine plastic waste [43];
9	2021	“3D Africa”	Youth for Technology Foundation, volunteers’ network	Global network to support the empowerment of African young people and women through STEM disciplines and 3D printing [44];
10	2021	“Rrreefs”	ETH Zürich	Startup for ecosystem restoration through 3D printed artificial coral reefs [45];

3.4 SDGs, key aspects, and practice-based case studies

The SDGs, the key aspects, and the case studies shown in the previous paragraphs were then combined in three different matrixes to further analyze the interconnections. In detail, the first matrix aims to highlight the links between the practice-based case studies and the key aspects from the preliminary literature review, while the second one is focused on the connection between the practice-based case studies and the 17 UN SDGs. After the previous matrixes focused on the case study analysis, the third one links the 8 key aspects with the 17 UN SDGs to detect some possible trends.

The first matrix is visible in Fig. 1 and resumes the connections between the selected case studies (rows) and the key aspects (columns). Four key aspects appear as more considered by the practitioners for more sustainable lifestyles, which are n. 1, 2, 3, and 6. As a matter of fact, personal fabrication and DIY practices represent a possible way to spread local actions in resilient communities, which can be globally connected from online networks, i.e., Appropedia [46]. These connections can be locally shared thanks to the makerspaces and fablabs. These can have an active role in sharing open knowledge and more sustainable ways of living through their initiatives. For instance, developing 3D printed assistive technology devices contributes to creating new products and knowledge that can be freely accessible thanks to open access repositories, allowing the distributed manufacturing of those products in different emerging countries or fragile contexts, as for case studies n. 2, 3, 4 and 7. Design projects involving AM can foster resilient local communities with efficient resource management, i.e. through ecosystem restoration, circular economy principles, and interspecies cooperation (case studies n. 5, 6, 7, 8, 10). Furthermore, designing for new knowledge and technology sharing can also facilitate new project collaborations amongst expert and non-expert users, inclusiveness and the fulfillment of specific user needs through mass customization, i.e., case studies n. 2, 3, and 9.

Key asp. Case Studies	1 Democratization	2 DIY	3 Local scales & communities	4 Assistive technology	5 Mass customization	6 Fablabs & Makerspaces	7 Circular Economy	8 Ecosystems
1 "Maker economy starter kit"	X	X	X		X	X		
2 "E-Nable"	X	X	X	X	X	X		
3 "One Hand for Syria"	X	X	X	X	X	X		
4 "Kijenzi"	X	X	X	X	X	X		
5 "H.O.R.T.U.S XL"		X	X			X		X
6 "Hiveopolis"	X	X	X			X	X	X
7 "Community First! Village"	X		X		X		X	
8 "The Elements"		X	X		X	X	X	X
9 "3D Africa"	X	X	X			X	X	
10 "Rrreefs"	X	X	X		X		X	X

Fig. 1. Data analysis matrix with the connections between the transdisciplinary key aspects (rows) and the practice-based case studies (columns) from this work.

Fig. 2 shows the second matrix with the connections between the selected case studies (rows) and the 17 UN SDGs (columns). As for the previous matrix, the three columns with the highest number of possible connections were highlighted to emphasize the SDGs that design practice and AM can mainly influence, which means SDG 10 (Reduced inequalities), SDG 11 (Sustainable cities and communities), and SDG 12 (Responsible consumption and production).

As can be noticed, the economic and social aspects show a higher number of possible connections concerning the environmental one, although the consequences of non-sustainable models often pass on to the Biosphere. However, the influence of human actions is more perceived in the Economy and Society. Indeed, they can be seen as embedded parts of the Biosphere, and concrete actions linked to the Biosphere are also connected with the social and economic aspects of sustainable development models [6, 7]. Local communities also play a crucial role in the transition towards sustainable models of consumption and production that can reduce inequalities amongst the citizens (case studies n. 1, 3, 4, and 7). These local actions can be achieved thanks to the collaboration amongst different stakeholders of a resilient community or through global exchange networks for ideas and knowledge sharing, i.e., case studies 2, 9, Appropedia platform [46]. These online collaborative environments can also promote the active implementation of the main best practices within different communities, encouraging their refinement and spread globally. Here, designers can help in developing these concrete strategies by using low-cost and accessible AM technologies as a tool for resilient and sustainable local communities.

UN SDGs	SDG 1	SDG 2	SDG 3	SDG 4	SDG 5	SDG 6	SDG 7	SDG 8	SDG 9	SDG 10	SDG 11	SDG 12	SDG 13	SDG 14	SDG 15	SDG 16	SDG 17
Case Studies																	
1 "Maker economy starter kit"	×			×				×		×	×	×	×				×
2 "E-Nable"	×		×		×					×	×	×	×				×
3 "One Hand for Syria"	×		×	×	×			×		×		×					×
4 "Kijensi"	×		×						×	×	×	×	×				
5 "H.O.R.T.U.S. XL"								×	×		×	×	×		×		
6 "Hiveopolis"		×						×	×		×		×		×		
7 "Community First! Village"	×		×					×	×	×	×	×	×				
8 "The Elements"						×		×	×	×	×	×		×			
9 "3D Africa"	×		×	×	×			×	×	×	×	×					×
10 "Rrreefs"				×	×	×		×	×	×	×	×	×	×	×		

Biosphere (environmental dimension)
 Society (social dimension)
 Economy (economic dimension)

SDGs: 1. No poverty // 2. Zero hunger // 3. Good health and well-being // 4. Quality education // 5. Gender equality // 6. Clean water and sanitation // 7. Affordable and clean energy // 8. Decent work and economic growth // 9. Industry, innovation and infrastructure // 10. Reduced inequalities // 11. Sustainable cities and communities // 12. Responsible consumption and production // 13. Climate action // 14. Life below water // 15. Life on land // 16. Peace, justice and strong institutions // 17. Partnerships for the goals

Fig. 2. Data analysis matrix with the connections between the practice-based case studies selected in this work (rows) and the UN 17 SDGs (columns).

3.5 Design practice and AM role in fostering SDGs

As shown in Fig. 3, the third matrix resumes the possible links between the key aspects (rows) and the 17 UN SDGs (columns), and it can be seen as a way to find some possible trends by resuming and generalizing the two previous two matrixes. In short, the key aspects of AM and design practice can be linked with the SDGs and the three aspects for sustainable development as it follows:

1. **Democratization of technology, open and frugal innovation** with the Biosphere (SDGs 13, 15), Society (SDGs 1, 2, 3, 4, 5, 11), Economy (SDGs 8, 9, 10, 12), and SDG 17.
2. **Personal Fabrication and Do-It-Yourself (DIY) practices** with the Biosphere (SDG 13), Society (SDGs 1, 3, 4, 5, 11), and Economy (SDGs 8, 10, 12).
3. **Distributed Manufacturing, local scales, and communities** with the Biosphere (SDGs 6, 13, 14, 15), Society (SDGs 1, 3, 11), and Economy (SDGs 8, 9, 12).
4. **Assistive Technology, Bioprinting, and Medicine** with the Society (SDG 3).
5. **Mass Customization** with the Society (SDGs 3, 11) and Economy (SDG 10).
6. **The spread of Fablabs and Makerspaces** with the Society (SDG 4, 11), Economy (SDG 8, 12), and SDG 17.

7. **Circular Economy enabled by AM** with the Biosphere (SDGs 6, 13, 14, 15), the Society (SDG 11), and Economy (SDGs 8, 9, 12).
8. **Ecosystem's restoration and preservation through AM** with the Biosphere (SDGs 6, 14, 15).

In other words, the key aspects 1, 2, 3, and 7 can be linked to the three aspects of sustainable development, which means the environmental, social, and economic ones. Biosphere mainly relates to: (i) personal fabrication and DIY practices, (ii) distributed manufacturing, local scales, and communities, and (iii) circular economy enabled by AM. Society is linked to: (i) democratization of technology, open and frugal innovation, and (ii) personal fabrication and DIY practices. Finally, Economy is more related to: (i) democratization of technology, open and frugal innovation, (ii) personal fabrication and DIY practices, (iii) distributed manufacturing, local scales, and communities, and (iv) circular economy enabled by AM. As for the second matrix, social and economic aspects count more possible connections. As a matter of fact, the columns of SDG 8 (Decent work and economic growth), SDG 11 (Sustainable cities and communities), and SDG 12 (Responsible consumption and production) show the highest numbers of links with the key aspects detected in this work.

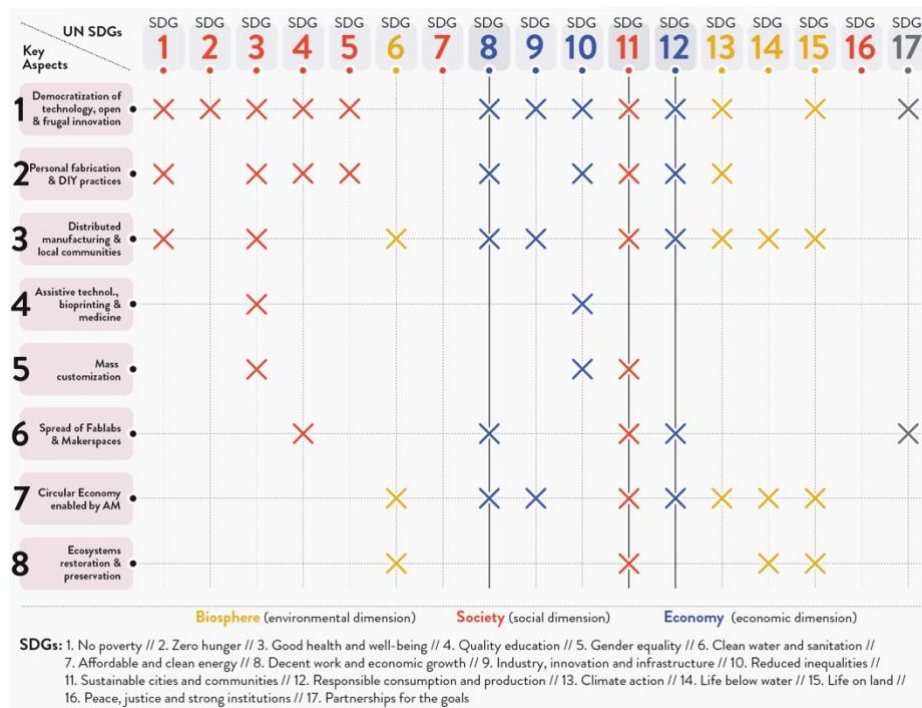


Fig. 3. Data analysis matrix with the connections between the transdisciplinary key aspects detected in this work (rows) and the 17 UN SDGs (columns).

The third matrix of Fig. 3 confirmed some trends considered during the analysis of the case studies, adding the contribution of circular economy practices in achieving the SDGs through design strategies enabled by AM. Resilient local communities and the global collaboration of different stakeholders can also contribute to spreading more sustainable ways of living. Furthermore, open innovation principles may foster this behavioral change in society, mainly thanks to the activities promoted by makerspaces and fablabs.

To sum up, AM technologies, remarkably affordable and low-cost 3D printing, can promote SDGs in several ways. From the analysis conducted in this study, the contribution of AM is almost 360 degrees, except perhaps only in the energy sector at an affordable price for everyone and in the context of policies (SDGs 7 and 16). As for the other points, AM technologies can contribute thanks significantly to their main characteristics: (i) being low cost; (ii) being flexible and able to respond to various problems; (iii) being consistent with the logic of open innovation, inclusiveness, and empowerment; (iv) giving tailor-made answers to different needs; and (v) being consistent with the principles of circularity.

The role of design practice in this sustainable transition is to make the know-how of the project available to find intelligent solutions that users can accept. The designer facilitates a change of behavior in users by making them accepted and desired the consistency of their actions with the SDGs. Hence, designers can be considered the potential facilitators for their achievement, and they can rely on AM technologies when and where needed, supporting resilient local communities.

4 Conclusions

To conclude, this paper aimed to connect AM and the design discipline by considering the environmental, social, and economic aspects of sustainability for new models of living. This review paper is particularly meant to link the 17 UN SDGs and AM through the lenses of the Design for Sustainability discipline, focusing on the roles of design and AM. Therefore, the main aspects of AM technologies and connections with the SDGs were detected thanks to the review of the most relevant case studies searched within the design practice. Accessible AM technologies can stimulate the development of a holistic approach to accomplish most of the UN SDGs, giving a concrete contribution to the transition toward more sustainable models of living. This can be done by defining several strategies based on the combination of the different key aspects explained before.

At the moment, particular interest was shown in spreading personal fabrication and DIY practices (Key aspect 2), developing distributed manufacturing strategies within local scales and communities (Key aspect 3), and creating global networks through Fablabs and Makerspaces (Key aspect 6). Reducing inequalities (SDG 10), creating sustainable cities and communities (SDG 11), and looking at responsible consumption and production ways (SDG 12) are the main sustainable development strategies that emerged from the case study analysis. Furthermore, reaching decent work and economic growth (SDG 8), sustainable cities and communities (SDG 11), and

responsible consumption and production (SDG 12) currently appear as the main sustainable development strategies enabled by AM and design practice especially thanks to open and frugal innovation (Key aspect 1), distributed manufacturing within local scales and communities (Key aspect 3), and circular economy principles i.e., Recycling, Reuse, Repair, Repurpose (Key aspect 7).

Currently, there is an increasing interest in AM technologies such as extrusion-based processes since they may be seen as accessible enablers for new projects and bottom-up or DIY initiatives, especially for local communities and developing countries. Furthermore, a significant role is represented by the development of new materials, which allows new strategies based on the recycling and reuse of waste and local resources for AM. Within this general context, designers can be considered facilitators who take advantage of AM only “on-demand”, which means just when and where needed. For these reasons, AM can be seen as a tool to foster a behavioral change toward more sustainable ways of living. Although some UN SDGs show more synergies with AM, other transdisciplinary strategies may be developed in the future to achieve all the SDGs, including SDG 7 (Affordable and Clean Energy) and SDG 16 (Peace, Justice, and Strong Institutions). Moreover, they would be used not only by the design practitioners but also by all the stakeholders actively involved in achieving the goals and fostering the sustainable transition, among which the end-users for self-productions and co-creation activities. Since this work is meant to give a preliminary overview of the possible interconnections with the UN SDGs, further work should be done to deepen the aspects detected by this work. Other repositories should be detected to widen the case study analysis, as well as some considerations, could be done considering different AM categories. As other projects will appear in the next decade, additional case studies can be detected in the next future, and they may potentially affect the current framework highlighted in this work.

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References

1. Lewis, S.L., Maslin, M.A.: Defining the Anthropocene. *Nature*. 519, 171–180 (2015). <https://doi.org/10.1038/nature14258>
2. Elhacham, E., Ben-Uri, L., Grozovski, J., Bar-On, Y.M., Milo, R.: Global human-made mass exceeds all living biomass. *Nature*. 588, 442–444 (2020). <https://doi.org/10.1038/s41586-020-3010-5>
3. Korhonen, J., Honkasalo, A., Seppälä, J.: Circular Economy: The Concept and its Limitations. *Ecol. Econ.* 143, 37–46 (2018). <https://doi.org/10.1016/j.ecolecon.2017.06.041>
4. Reike, D., Vermeulen, W.J.V., Witjes, S.: The circular economy: New or Refurbished as CE 3.0? — Exploring Controversies in the Conceptualization of the Circular Economy through a Focus on History and Resource Value Retention Options. *Resour. Conserv. Recycl.* 135, 246–264 (2018). <https://doi.org/10.1016/j.resconrec.2017.08.027>

5. Sauv  , S., Bernard, S., Sloan, P.: Environmental sciences, sustainable development and circular economy: Alternative concepts for trans-disciplinary research. *Environ. Dev.* 17, 48–56 (2016). <https://doi.org/10.1016/j.envdev.2015.09.002>
6. Stockholm Resilience Centre SRC: Contributions to Agenda 2030, <https://www.stockholmresilience.org/research/research-news/2017-02-28-contributions-to-agenda-2030.html>, last accessed 2022/03/11.
7. United Nations: The 17 Goals, <https://sdgs.un.org/goals>, last accessed 2022/05/19.
8. Camocho, D., Ferreira, A.M., Vicente, J.: TRANSition to circular and sustainable economy through design. In: Proceedings of the 5th Design Doctoral Conference, DDC'18: Transgression. IADE - Universidade Europeia. 8 (2018)
9. Gaziulusoy, A.  ., Erdo  an   ztekin, E.: Design as a Catalyst for Sustainability Transitions. In: DRS 2018: Catalyst. 1041–1051 (2018)
10. Dantas, T.E.T., de-Souza, E.D., Destro, I.R., Hammes, G., Rodriguez, C.M.T., Soares, S.R.: How the combination of Circular Economy and Industry 4.0 can contribute towards achieving the Sustainable Development Goals. *Sustain. Prod. Consum.* 26, 213–227 (2021). <https://doi.org/10.1016/j.spc.2020.10.005>
11. Berawi, M.A.: The Role of Industry 4.0 in Achieving Sustainable Development Goals. *Int. J. Technol.* 10, 644 (2019). <https://doi.org/10.14716/ijtech.v10i4.3341>
12. Schroeder, P., Anggraeni, K., Weber, U.: The Relevance of Circular Economy Practices to the Sustainable Development Goals. *J. Ind. Ecol.* 23, 77–95 (2019). <https://doi.org/10.1111/jiec.12732>
13. Agrawal, R., S., V.: State of art review on sustainable additive manufacturing. *Rapid Prototyp. J.* 25, 1045–1060 (2019). <https://doi.org/10.1108/RPJ-04-2018-0085>
14. Attaran, M.: The rise of 3-D printing: The advantages of additive manufacturing over traditional manufacturing. *Bus. Horiz.* 60, 677–688 (2017). <https://doi.org/10.1016/j.bushor.2017.05.011>
15. Despeisse, M., Baumers, M., Brown, P., Charnley, F., Ford, S.J., Garmulewicz, A., Knowles, S., Minshall, T.H.W., Mortara, L., Reed-Tsochas, F.P., Rowley, J.: Unlocking value for a circular economy through 3D printing: A research agenda. *Technol. Forecast. Soc. Change.* 115, 75–84 (2017). <https://doi.org/10.1016/j.techfore.2016.09.021>
16. Dressen, K., Schepers, S., Leen, D.: From hacking things to making things. Rethinking making by supporting non-expert users in a FabLab. *Interact. Des. Archit.* 30, 47–64 (2016)
17. Rayna, T., Striukova, L.: From rapid prototyping to home fabrication: How 3D printing is changing business model innovation. *Technol. Forecast. Soc. Change.* 102, 214–224 (2016). <https://doi.org/10.1016/j.techfore.2015.07.023>
18. Diegel, O., Singamneni, S., Reay, S., Withell, A.: Tools for Sustainable Product Design: Additive Manufacturing. *J. Sustain. Dev.* 3, 68 (2010). <https://doi.org/10.5539/jsd.v3n3p68>
19. Sauerwein, M., Doubrovski, E., Balkenende, R., Bakker, C.: Exploring the potential of additive manufacturing for product design in a circular economy. *J. Clean. Prod.* 226, 1138–1149 (2019). <https://doi.org/10.1016/j.jclepro.2019.04.108>
20. Cl  ries, L., Rognoli, V.: Materials Designers: A New Design Discipline. In: Material Designers - Boosting talent towards circular economies. 43–47 (2021)

21. Rognoli, V., Bianchini, M., Maffei, S., Karana, E.: DIY materials. *Mater. Des.* 86, 692–702 (2015). <https://doi.org/10.1016/j.matdes.2015.07.020>
22. Romani, A., Rognoli, V., Levi, M.: Design, Materials, and Extrusion-Based Additive Manufacturing in Circular Economy Contexts: From Waste to New Products. *Sustainability*. 13, 7269 (2021). <https://doi.org/10.3390/su13137269>
23. Albert, M.: Sustainable frugal innovation - The connection between frugal innovation and sustainability. *J. Clean. Prod.* 237, 117747 (2019). <https://doi.org/10.1016/j.jclepro.2019.117747>
24. Boisseau, É., Omhover, J.-F., Bouchard, C.: Open-design: A state of the art review. *Des. Sci.* 4, e3 (2018). <https://doi.org/10.1017/dsj.2017.25>
25. Gibson, I., Shukla, A.: Sustainable Frugal Design Using 3D Printing. In: Muthu, S.S. and Savalani, M.M. (eds.) *Handbook of Sustainability in Additive Manufacturing*. 85–100. Springer Singapore, Singapore (2016)
26. Kreiger, M., Pearce, J.M.: Environmental Life Cycle Analysis of Distributed Three-Dimensional Printing and Conventional Manufacturing of Polymer Products. *ACS Sustain. Chem. Eng.* 1, 1511–1519 (2013). <https://doi.org/10.1021/sc400093k>
27. Prendeville, S., Hartung, G., Purvis, E., Brass, C., Hall, A.: Makespaces: From Redistributed Manufacturing to a Circular Economy. In: Setchi, R., Howlett, R.J., Liu, Y., and Theobald, P. (eds.) *Sustainable Design and Manufacturing 2016*. 577–588. Springer International Publishing, Cham (2016)
28. Laplume, A.O., Petersen, B., Pearce, J.M.: Global value chains from a 3D printing perspective. *J. Int. Bus. Stud.* 47, 595–609 (2016). <https://doi.org/10.1057/jibs.2015.47>
29. Buehler, E., Branham, S., Ali, A., Chang, J.J., Hofmann, M.K., Hurst, A., Kane, S.K.: Sharing is Caring: Assistive Technology Designs on Thingiverse. In: *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. pp. 525–534. Association for Computing Machinery, New York, NY, USA (2015)
30. Da Silveira, G., Borenstein, D., Fogliatto, F.S.: Mass customization: Literature review and research directions. *Int. J. Prod. Econ.* 72, 1–13 (2001). [https://doi.org/10.1016/S0925-5273\(00\)00079-7](https://doi.org/10.1016/S0925-5273(00)00079-7)
31. Mourtzis, D.: Design of customised products and manufacturing networks: towards frugal innovation. *Int. J. Comput. Integr. Manuf.* 31, 1161–1173 (2018). <https://doi.org/10.1080/0951192X.2018.1509131>
32. Mohammed, J.S.: Applications of 3D printing technologies in oceanography. *Methods Oceanogr.* 17, 97–117 (2016). <https://doi.org/10.1016/j.mio.2016.08.001>
33. Walker, M., Humphries, S.: 3D Printing: Applications in evolution and ecology. *Ecol. Evol.* 9, 4289–4301 (2019). <https://doi.org/10.1002/ece3.5050>
34. Suárez, L., Domínguez, M.: Sustainability and environmental impact of fused deposition modelling (FDM) technologies. *Int. J. Adv. Manuf. Technol.* 106, 1267–1279 (2020). <https://doi.org/10.1007/s00170-019-04676-0>
35. ASTM International: ISO/ASTM 52900-15 Standard Terminology for Additive Manufacturing – General Principles – Terminology. ASTM International, West Conshohocken, PA (2015)
36. Maker Economy Starter Kit., <https://www.3dwasp.com/maker-economy-starter-kit/#more>, last accessed 2021/11/26.

37. E-nable., <https://enablingthefuture.org/e-nable-community-chapters/>, last accessed 2021/11/26.
38. One hand for Syria., <https://www.3dwasp.com/wasp-protesi-3d-siria/>, last accessed 2021/11/26.
39. Development of a Resilient 3-D Printer for Humanitarian Crisis Response, https://www.appropedia.org/Development_of_a_Resilient_3-D_Printer_for_Humanitarian_Crisis_Response, last accessed 2021/11/26.
40. H.O.R.T.U.S. XL., <https://www.ecologicstudio.com>, last accessed 2021/11/26.
41. HIVEOPOLIS - beehives of the future., <https://www.hiveopolis.eu/>, last accessed 2021/11/26.
42. ICON: ICON Delivers Series of 3D-Printed Homes for Homeless in Austin | ICON, <https://www.iconbuild.com/updates/icon-delivers-series-of-3d-printed-homes-for-homeless>, last accessed 2021/11/26.
43. Elements - The New Raw., <https://thenewraw.org/Elements>, last accessed 2021/11/26.
44. 3D Africa., <https://3dafrica.org/>, last accessed 2021/11/26.
45. rreefs., <https://www.rreefs.com>, last accessed 2021/11/26.
46. Welcome to Appropedia, https://www.appropedia.org/Welcome_to_Appropedia, last accessed 2021/11/26.